14/03/24 MS

USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] **[23TCSE202]**

SECOND SEMESTER B.Tech EXAMINATION, MARCH-2024

PYTHON PROGRAMMING

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i.** Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

PART-A **5 X 2=10**

1.	**a.**	Differentiate between mutable and immutable objects in Python language.	L-2	CO1	**02**
	b.	What are the basic data structures used in Python programming?	L-2	CO2	**02**
	c.	Explain class constructor is in Python.	L-2	CO2	**02**
	d.	Describe inheritance in Python.	L-2	CO3	**02**
	e.	Explain matrix. How it is represented in Python?	L-2	CO4	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Explain the difference between a for loop and a while loop in Python.	L-2	CO1	**05**
		OR			
	b.	Demonstrate the use of the break and continue statements within nested loops.	L-2	CO1	**05**
3.	**a.**	List the reasons for defining functions.	L-2	CO2	**05**
		OR			
	b.	Explain the concept of variable number of arguments in Python functions.	L-3	CO2	**05**
4.	**a.**	Explain the concept of decorators in Python and how they are used to modify the behavior of functions.	L-2	CO2	**05**
		OR			
	b.	Explain the concept of class customization in Python.	L-2	CO3	**05**
5.	**a.**	Explain the following concepts related to NumPy in Python: **i.** Counters **ii.** Random number generation	L-2	CO5	**05**
		OR			

	b.	Explain the following concepts related to NumPy in Python with one example: i. Creating arrays ii. Indexing and slicing arrays	L-2	CO5	**05**
6.	**a.**	Explain the following concepts related to file handling and binary data in Python: i. Getting files from the Internet ii. Reading files iii. Writing files	L-2	CO4	**05**
		OR			
	b.	Explain method overriding with an example.	L-2	CO4	**05**
7.	**a.**	Explain indexing and slicing in python.	L-2	CO5	**05**
		OR			
	b.	Explain with an example the concepts of membership and identity operators in Python.	L-2	CO5	**05**
		PART-C		**6 X 10=60**	
8.	**a.**	Demonstrate break and continue with an example.	L-3	CO1	**10**
		OR			
	b.	Explain the concept of conditional statements, equality and relational operators in Python.	L-3	CO1	**10**
9.	**a.**	Explain with code snippet the importance of search algorithms (linear and binary search) in programming.	L-3	CO2	**10**
		OR			
	b.	Explain the concept of recursion in Python.	L-3	CO2	**10**
10.	**a.**	Explain the concept of handling exceptions using try and except blocks in Python, including the use of multiple exception handlers.	L-2	CO3	**10**
		OR			
	b.	Explain with an example the concept of exceptions within functions, the use of finally to clean up resources, and the creation of custom exception types in Python.	L-2	CO3	**10**

11.	**a.**	Explain with example the following concepts in Python programming: Comma separated values (CSV) files, and overriding class methods.	L-3	CO4	**10**
		OR			
	b.	Differentiate between "is-a" and "has-a" relationships in object-oriented programming with an example for each.	L-3	CO4	**10**
12.	**a.**	Explain the concept of performing operations on sparse matrices using SciPy in Python.	L-3	CO5	**10**
		OR			
	b.	Distinguish between mixing classes and multiple inheritance.	L-3	CO5	**10**
13.	**a.**	Explain the concepts of mutability and references in Python.	L-3	CO6	**10**
		OR			
	b.	Discuss the difference between class methods and instance methods.	L-3	CO6	**10**